VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi-590018, Karnataka,India

A Case Study Report on

"Elementary Conservation of Water and Subsurface Investigation of Groundwater"

Submitted in partial fulfilment of the Case Study requirement for sixth semester of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE & ENGINEERING

Submitted by

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Approved by AICTE,Govt. of India, New Delhi.

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2025-2026

CERTIFICATE

This is to certify that case study work entitled **“Elementary Conservation of Water and Subsurface Investigation of Groundwatert”** carried out by **Kavana C (1TJ23IC025)** in partial fulfilment for **sixth** semester of **Bachelor of Engineering in Computer Science** of **Visvesvaraya Technological University, Belagavi** during the academic year **2025-2026**. It is certified that all the corrections/suggestions indicated for the internal assessment have been incorporated in the report deposited in the departmental library. The case study report has been approves as it satisfies the academic requirements in respect of case study work as prescribed for the said degree.

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DECLARATION

I, Kavana C (1TJ23IC025), hereby declare that the case study entitled **“Elementary Conservation of Water and Subsurface Investigation of Groundwater”,** has been carried out and submitted by me as partial fulfilment of **sixth semester** of **Bachelor of Engineering in Computer Science and Engineering, Visvesvaraya Technological University (VTU),** during the academic year **2025-2026.** I also declare that, to the best of my knowledge and belief, the work reported here is accepted and satisfied.

Kavana C (1TJ23IC025)

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The case study report on "**Elementary Conservation of Water and Subsurface Investigation of Groundwater"** is the outcome of guidance, moral support and knowledge imparted on me, throughout my work. For this I acknowledge and express immense gratitude to all those who have guided and supported me during the preparation of this case study.

I take this opportunity to express my gratefulness to everyone who has extended their support for helping me in the case study completion.

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Kavana C(1TJ23IC025)

ABSTRACT

Groundwater is a vital natural resource that supports domestic, agricultural, and industrial activities, especially in regions facing water scarcity. Subsurface investigation of groundwater involves the study of underground water systems to determine their availability, distribution, movement, and quality. Techniques such as borehole drilling, geophysical surveys, and water table analysis are commonly used to identify aquifers and evaluate their potential for sustainable use. These investigations play a key role in effective planning and management of water resources.

However, increasing demand, over-extraction, and environmental changes have led to a decline in groundwater levels and deterioration in water quality. This highlights the need for proper conservation measures to maintain a balance between usage and replenishment. Elementary water conservation practices, such as rainwater harvesting, efficient irrigation methods, and reuse of wastewater, help in reducing water wastage and enhancing groundwater recharge.

This study emphasizes the importance of combining scientific investigation with basic conservation techniques to ensure the sustainable use of groundwater resources. Proper management and awareness can help protect this valuable resource and secure water availability for future generations.

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